
WORKSHOP #1 - FUNDAMENTALS



Presenter

Abid Nasim



Topic

AI Soup to Nuts

SESSION 1 OF 4 • TWO HOURS

AI Fundamentals

Background Knowledge You Need Before We Think Critically About AI

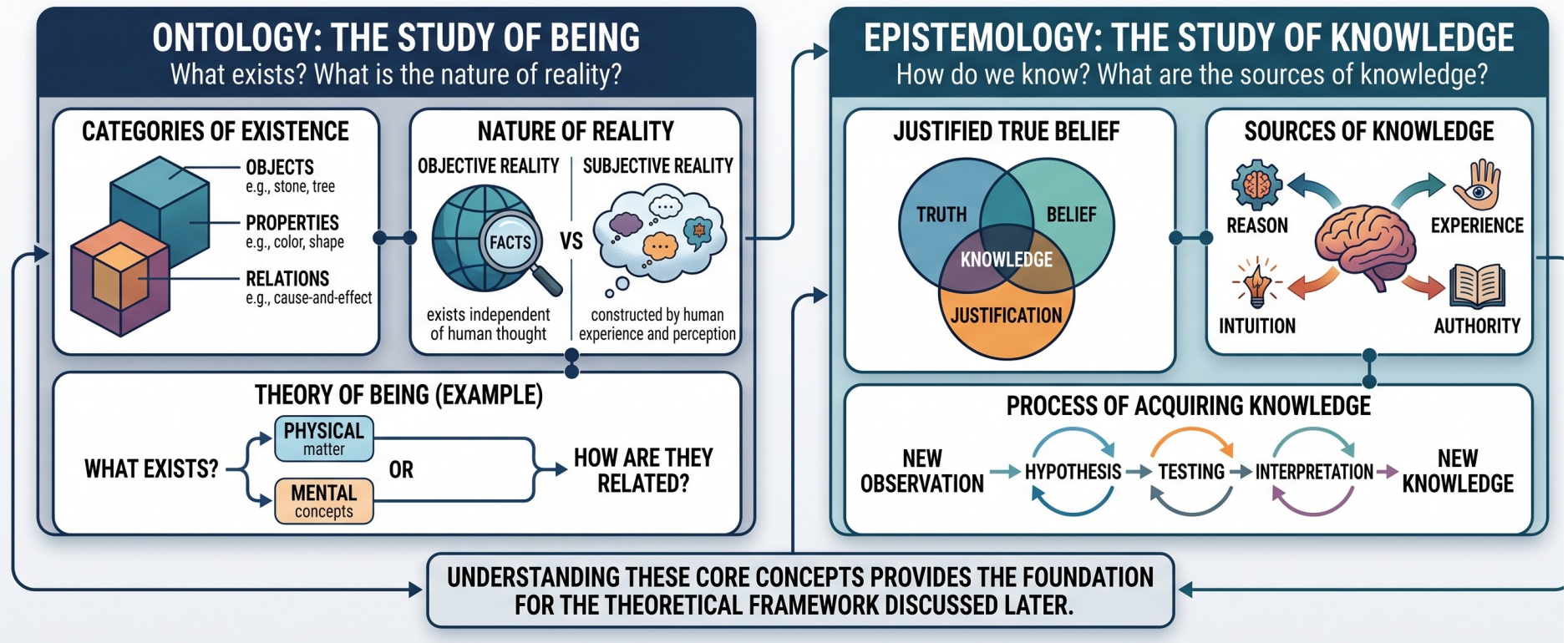
Where AI came from, the vocabulary it runs on, how it actually works under the hood, and what it can and can't do today.

This session is the foundation for “How to Think About AI” (Session 2).



PRESENTATION FRAMEWORK: A PHILOSOPHICAL DISCLAIMER

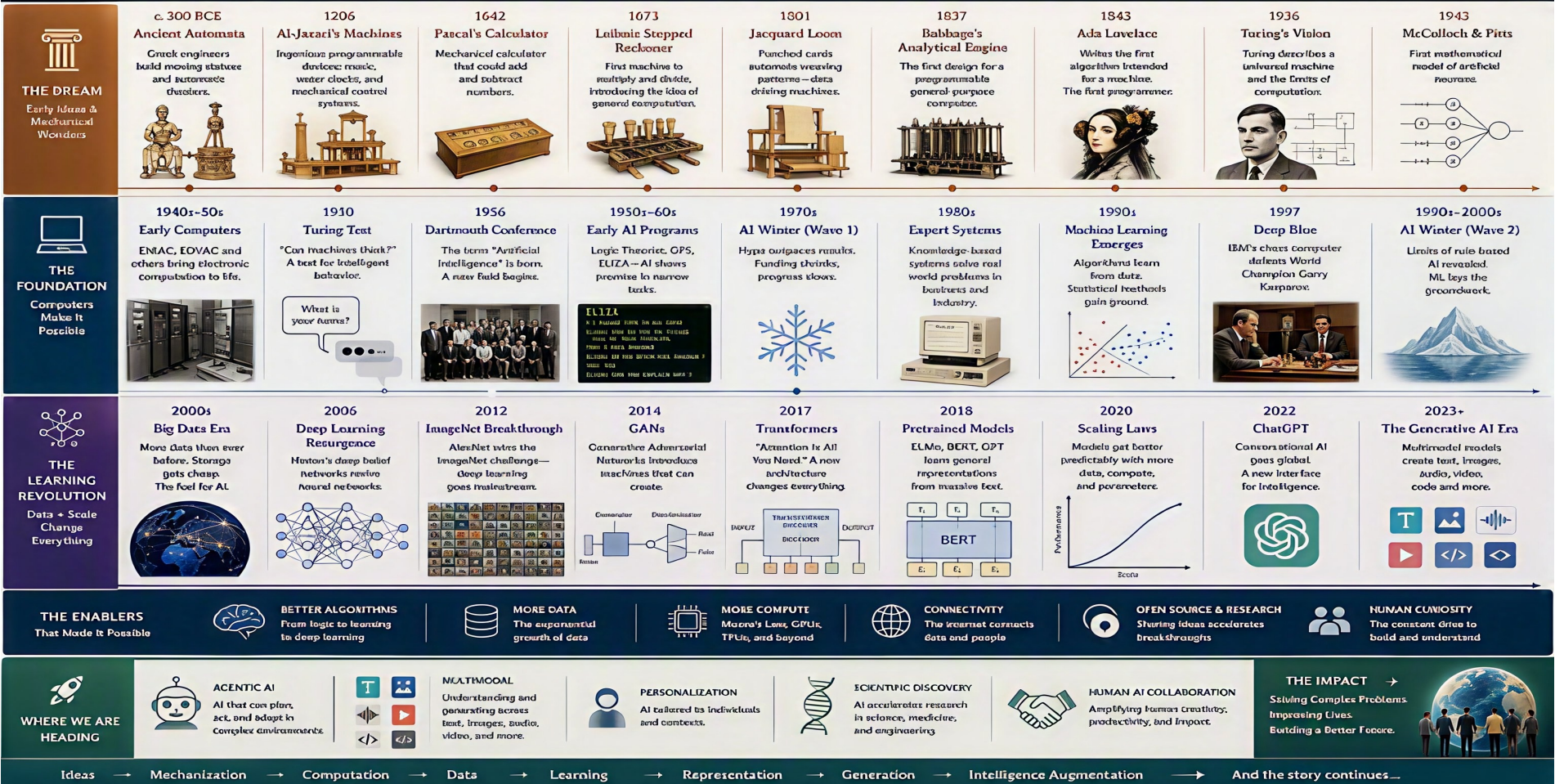
PLEASE BE ADVISED: This presentation will involve **THEORETICAL DISCUSSION**, specifically concerning fundamental philosophical concepts.



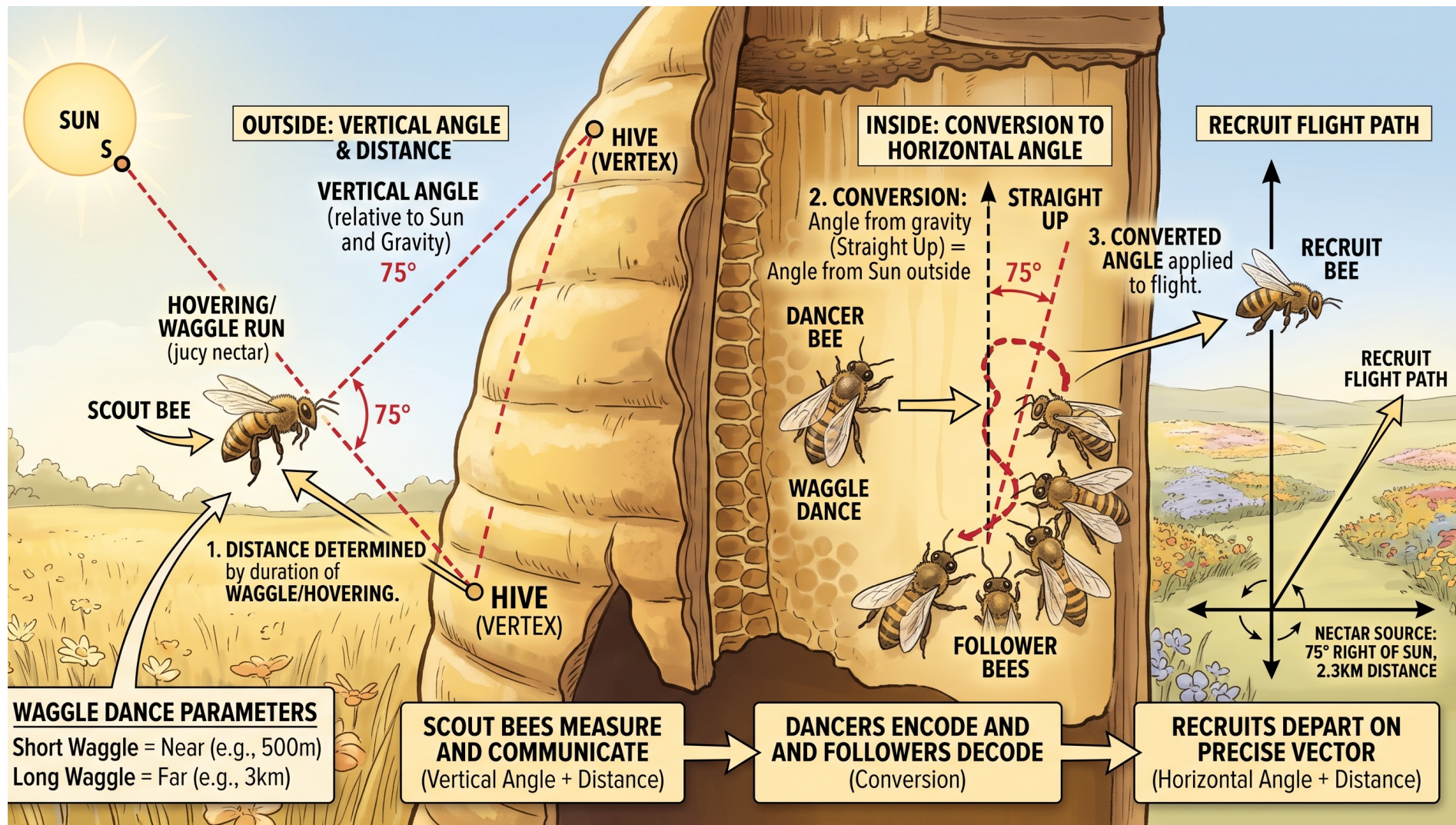
THE EVOLUTION OF AI

FROM ANCIENT AUTOMATA TO MODERN GENERATIVE INTELLIGENCE

A STORY OF HUMAN CURIOSITY,
TOOLS, IDEAS, AND COMPUTATION
COMING TOGETHER.

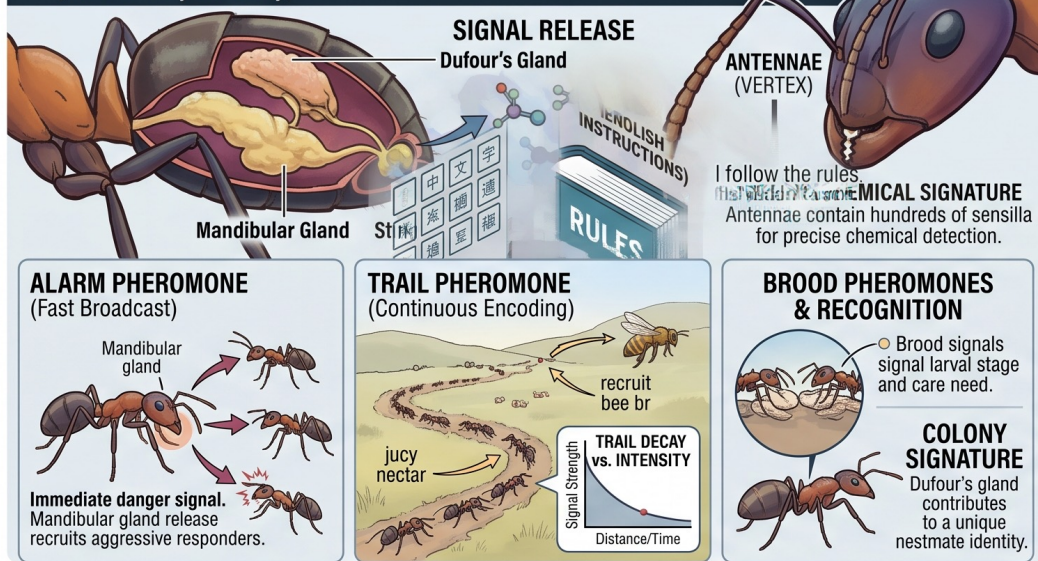


Ideas → Mechanization → Computation → Data → Learning → Representation → Generation → Intelligence Augmentation → And the story continues...

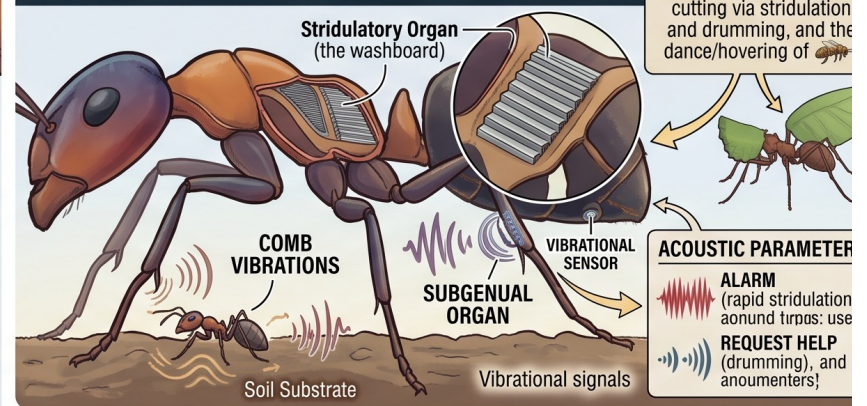


THE COMPLEXITY OF ANT COMMUNICATION: A MULTI-MODAL SYSTEM

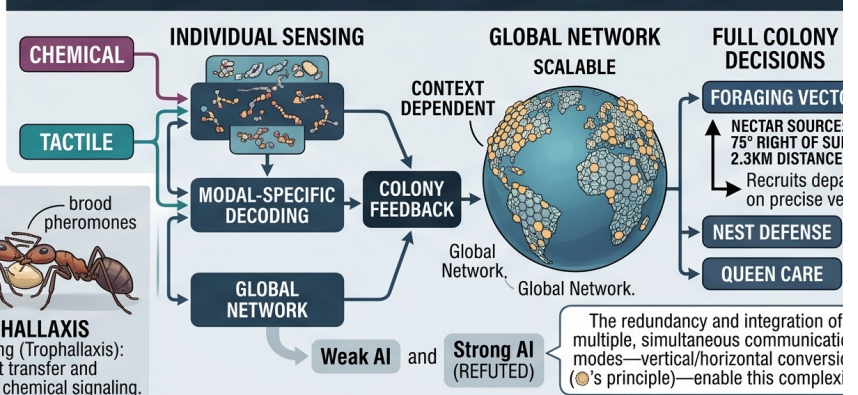
1. CHEMICAL COMMUNICATION: THE PHEROMONE LANGUAGE (The Primary Modality)



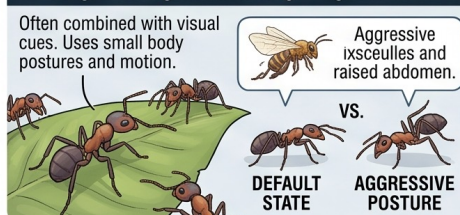
4. ACOUSTIC & VIBRATIONAL SIGNALS: THE SUBSTRATE-BORN COMMUNICATION



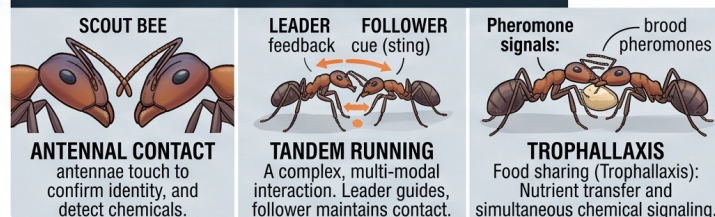
5. INTEGRATED SYSTEM: COMPLEX DECISION-MAKING



2. VISUAL COMMUNICATION: Augmenting Close-Range Signals

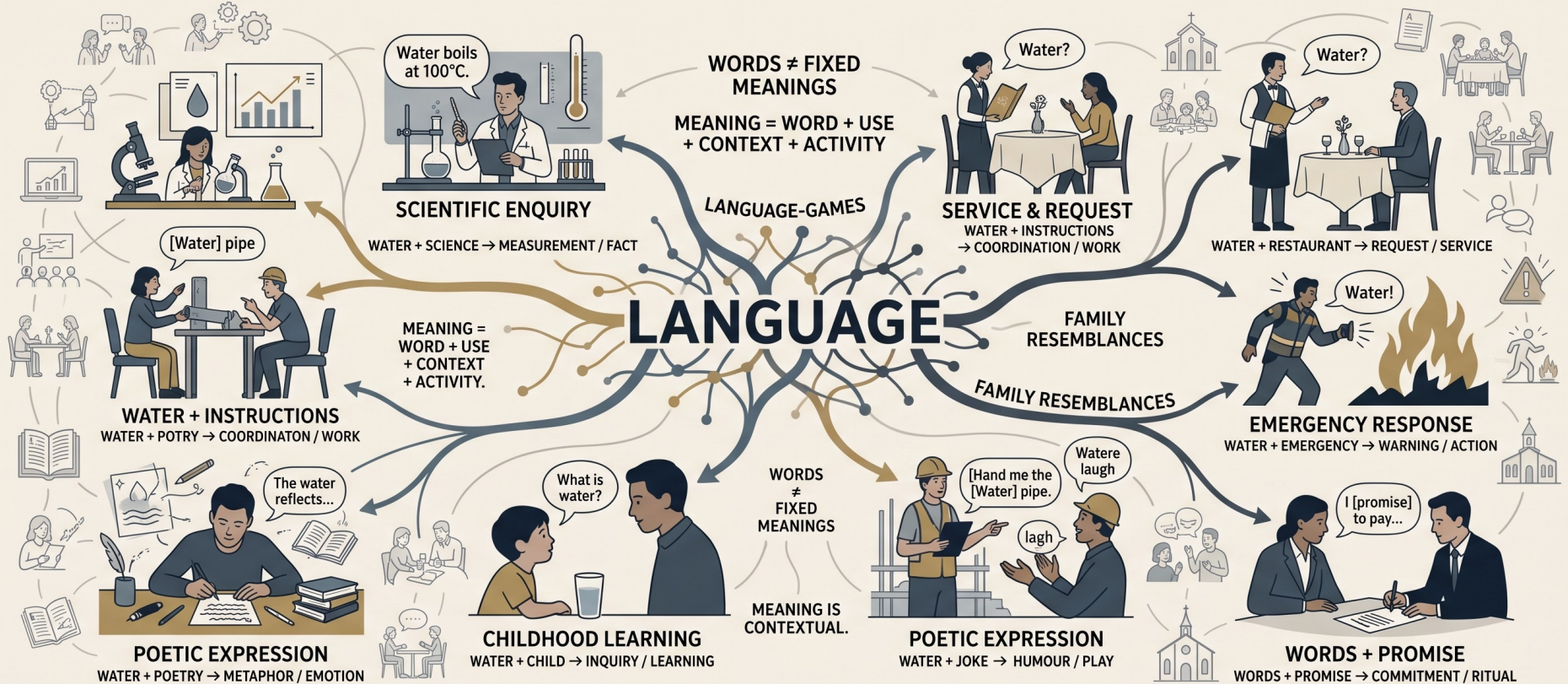


3. TACTILE COMMUNICATION: CLOSE-RANGE INTERACTIONS



WITTGENSTEIN'S LATER PHILOSOPHY OF LANGUAGE

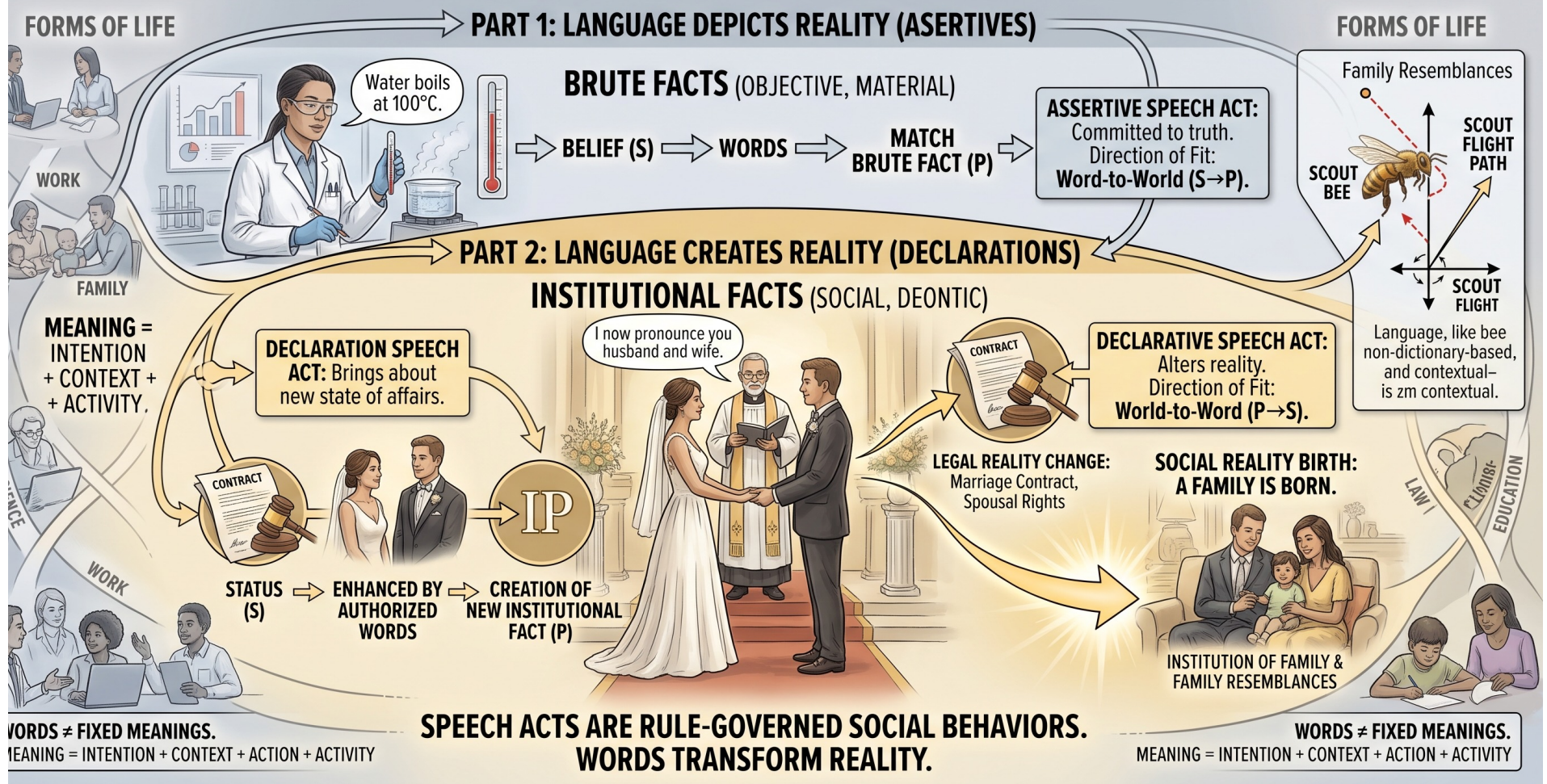
EXPLORING PHILOSOPHICAL INVESTIGATIONS: THE MEANING IS THE USE



MEANING IS NOT SIMPLY INSIDE THE WORD.

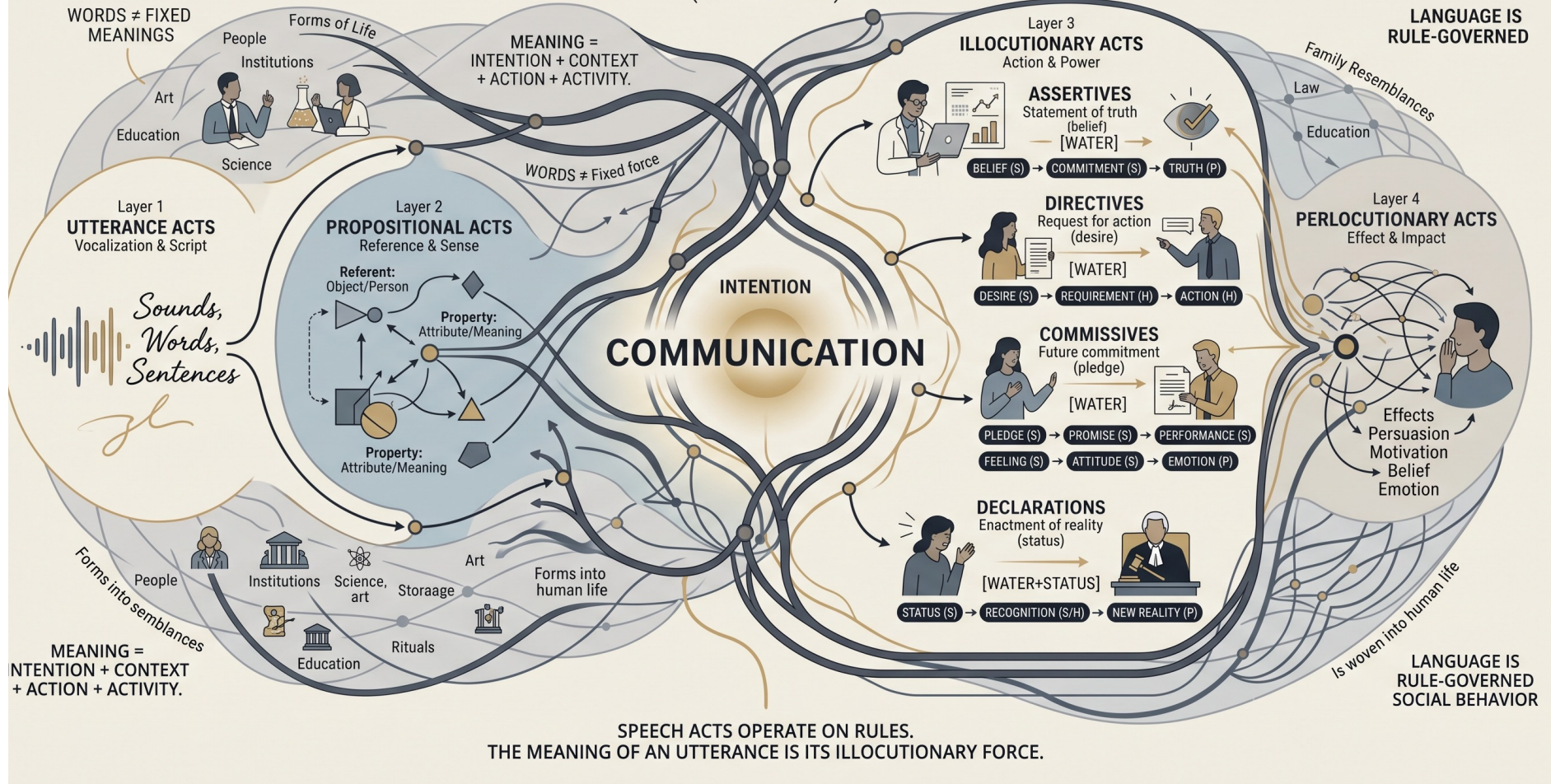
MEANING IS NOT SIMPLY INSIDE THE WORD. MEANING EMERGES FROM HOW THE WORD IS USED.

THE FUNCTION OF LANGUAGE: DEPICTING REALITY VS. CREATING SOCIAL REALITY (SPEECH ACTS)



THE ARCHITECTURE OF COMPLEX SPEECH ACTS

(AFTER SEARLE)



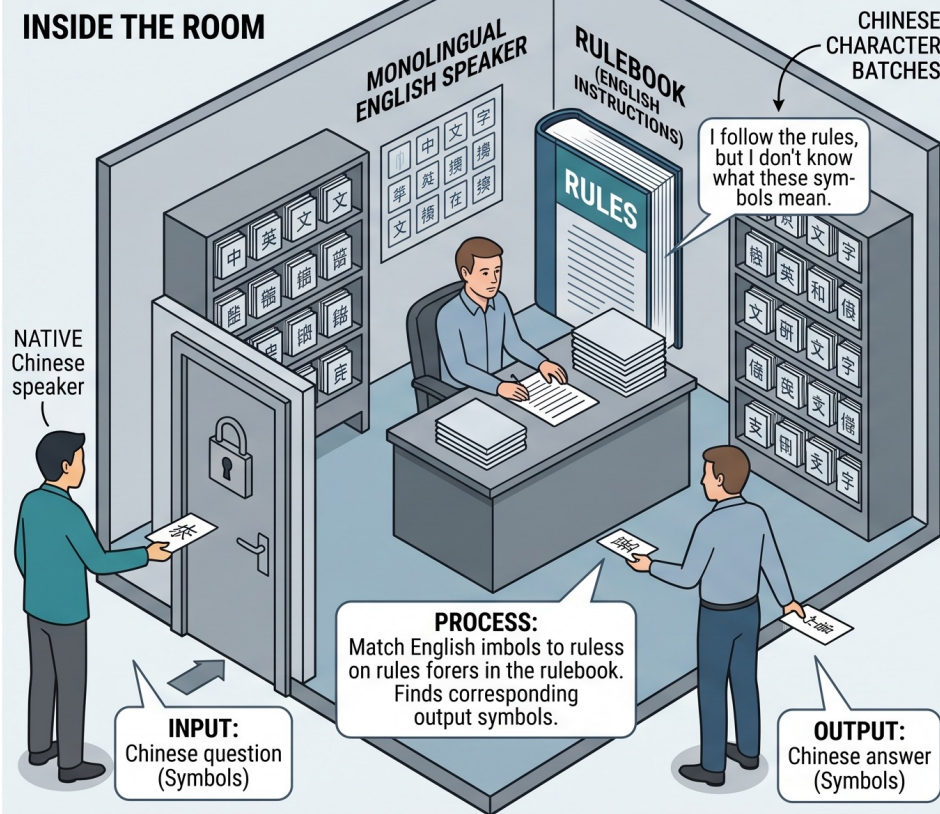
JOHN SEARLE'S CHINESE ROOM: THOUGHT EXPERIMENT OVERVIEW



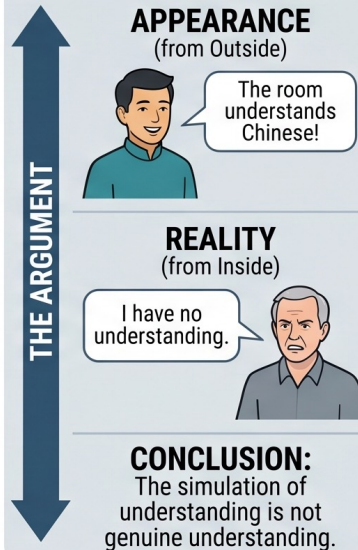
SYNTAX IS NOT SUFFICIENT FOR SEMANTICS.

1. THE ROOM (THE EXPERIMENT)

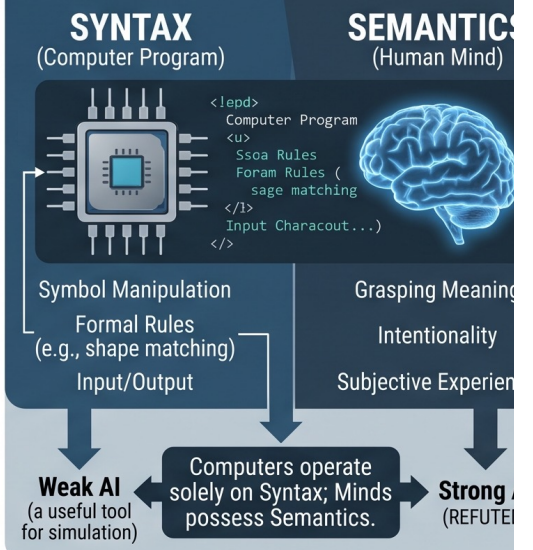
INSIDE THE ROOM



2. THE CONCLUSION (MERE SIMULATION)



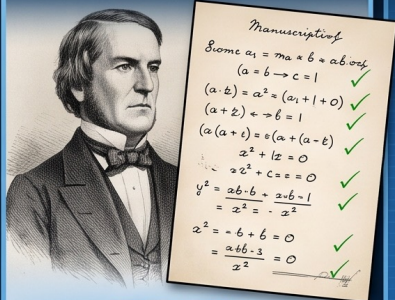
3. SYNTAX VS. SEMANTICS (A FUNDAMENTAL DISTINCTION)



MAJOR REPLIES & CRITIQUES

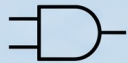
1. SYSTEMS REPLY:	2. ROBOT REPLY:	3. BRAIN SIMULATOR REPLY:	4. THE 'CRITIQUE OF THE CRITIQUE'
The entire system (person + room + rulebook) understands Chinese.	Embed the room in a robot with sensory input to give it "real" understanding.	Simulate the exact neural networks of a human brain to produce consciousness.	Can syntax, at a sufficient level of complexity, give rise to semantics?

1. BOOLEAN ALGEBRA (Mathematics of Logic)



OPERATIONS

TRUE (1) / FALSE (0)



AND (Multiplication)

A	B	D	Y
0	1	0	0
0	0	1	1
1	1	0	0



OR (Addition)

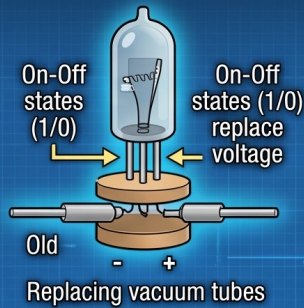
A	B	D	Y
0	1	0	0
0	0	1	1
1	0	0	1



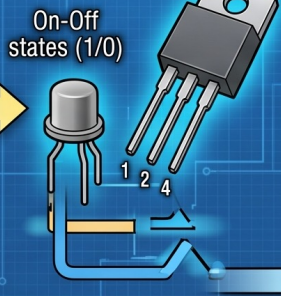
NOT (Inversion)

A	B	D	Y
0	1	0	0
0	0	1	1
1	1	0	0

2. TRANSISTOR (Electronic Switch)



TRANSLATION



3. LOGIC GATES (Building Blocks)

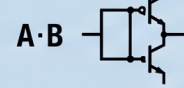
NOT

A	B	Y
A	0	0
0	1	Y



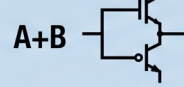
AND

A	B	Y
A	0	0
0	1	Y
0	0	Y



NAND

A	B	Y
A	0	0
0	1	Y
0	0	Y



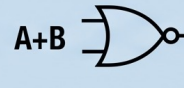
OR

A	B	Y
A	0	0
0	1	Y
0	1	Y



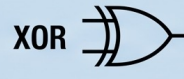
NOR

A	B	Y
A	0	0
0	0	Y
0	1	Y

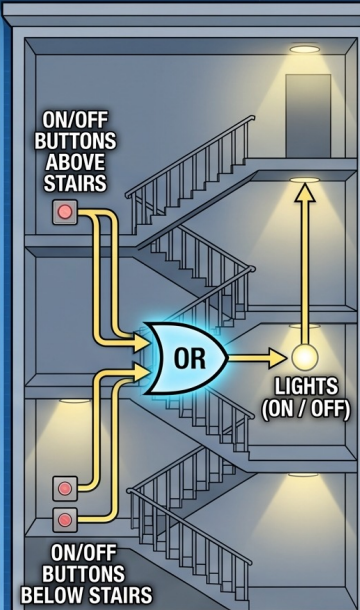


XOR

A	B	Y
A	0	Y
1	1	Y



4. OR GATE APPLICATION: STAIRWELL LIGHTING

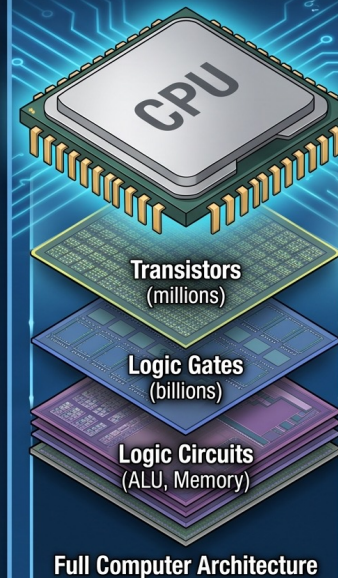


ANY switch can activate the light.

A || B || C || D = Light

The redundant OR gate principle, for techstants the souses to the conilunt of the redundant OR gate.

5. COMPUTER & MICROPROCESSOR



10110...
10110...
10110...
Boole & Transistor

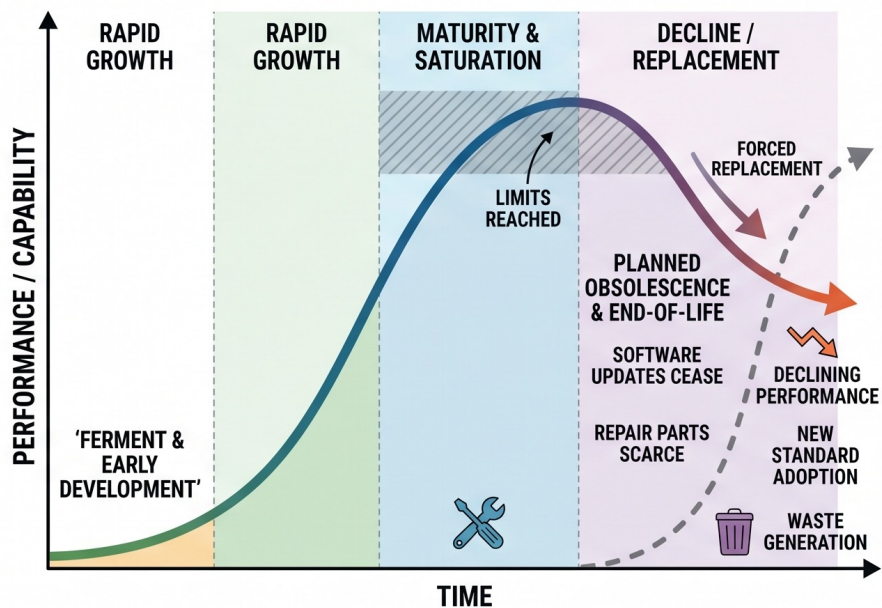
Computation



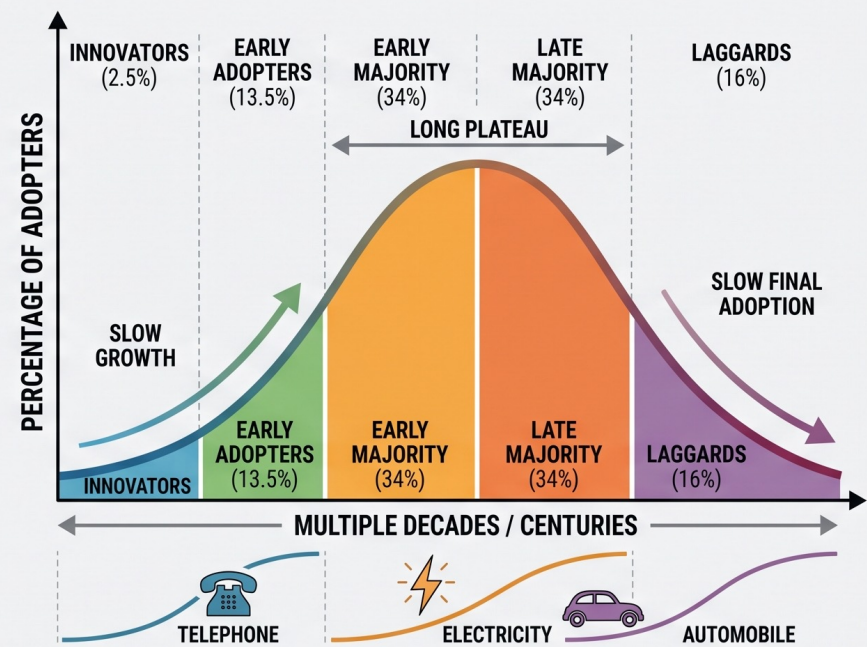
Computatio

EVOLUTION OF COMPUTATION

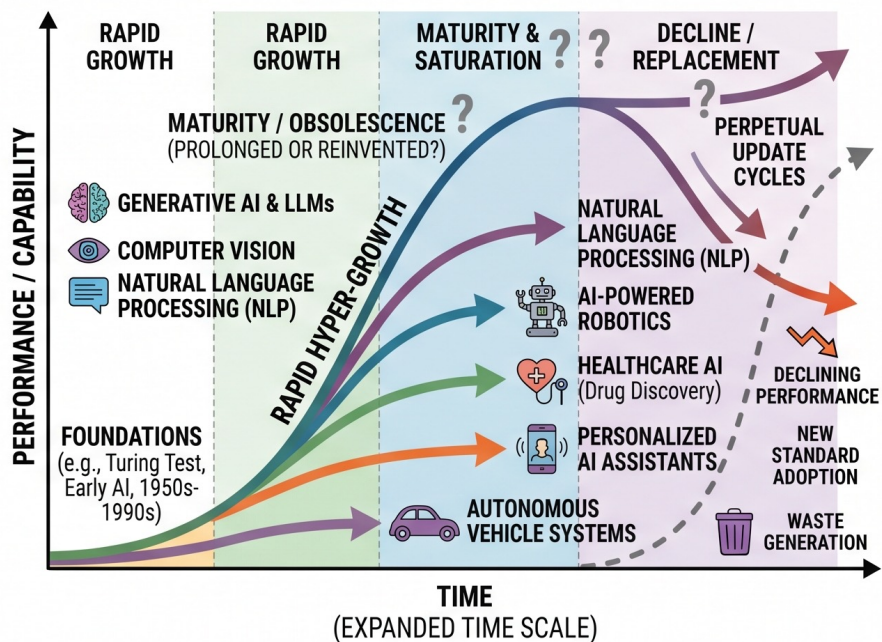
TECHNOLOGY S-CURVE: MATURITY & PLANNED OBSOLESCENCE



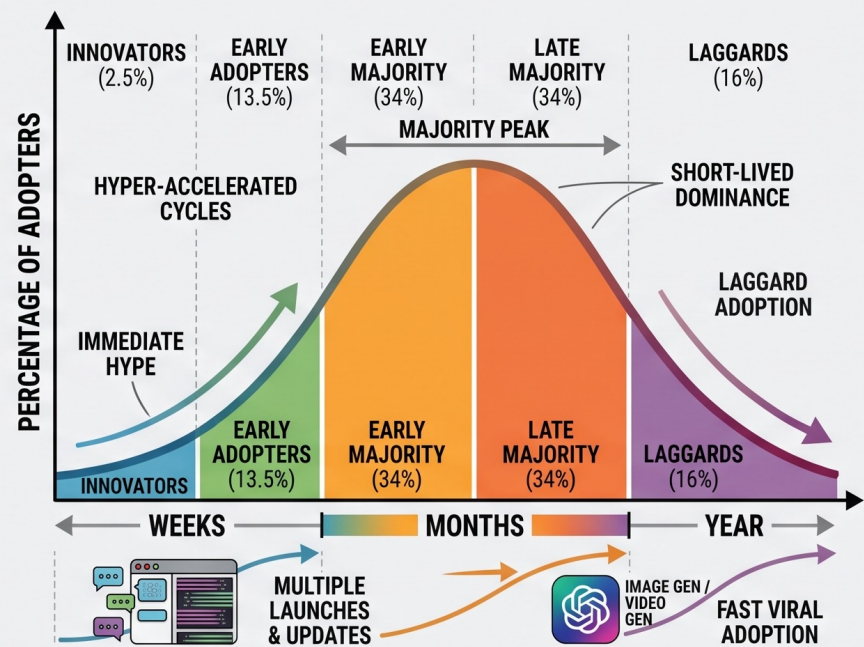
TECHNOLOGY ADOPTION CURVE (LONG-TERM HISTORICAL VIEW)



AI TECHNOLOGY S-CURVE: HYPER-GROWTH & OFFSHOOTS



AI TECHNOLOGY ADOPTION CURVE RAPID ADOPTION CYCLE (MONTHS to YEARS)



PART 1 • HISTORY

A (very) condensed history of AI

1956

Dartmouth Conference coins the term “Artificial Intelligence”

1958

Rosenblatt's Perceptron — the first trainable neural network

1970s & 90s

“AI winters” as early hype outpaces results

1986

Backpropagation popularized by Rumelhart, Hinton & Williams

2006

Hinton's deep belief networks spark the deep learning resurgence

2012

AlexNet wins ImageNet — deep CNNs prove they scale.

2017

“Attention Is All You Need” introduces the Transformer

2022

ChatGPT & generative AI reach everyday use

2024

Hinton & Hopfield win the Nobel Prize in Physics

A FRAMEWORK FOR AI SYSTEM DESIGN

The Periodic Table of AI Components

Columns group components by function; rows track how each matures from raw primitive to emerging practice.

	REACTIVE	RETRIEVAL	ORCHESTRATION	VALIDATION	MODEL
Primitives	1 Pr Prompting	2 Em Embeddings	3 Fn Function Calling	4 Gd Guardrails	5 Wt Model Weights
Compositions	6 Ch Chat Memory	7 Rg RAG Pipeline	8 Ac Agent Chains	9 Ev Eval Harness	10 Ft Fine-Tuning
Deployment	11 St Streaming	12 Vs Vector Store	13 Wf Workflow Engine	14 Ob Observability	15 Is Inference Serving
Emerging	16 Ea Event-Driven Agents	17 Gr Graph RAG	18 Ma Multi-Agent Systems	19 Sc Self-Critique	20 Mx Mixture of Experts

GLOSSARY

Eight terms worth knowing cold

Model

A mathematical function whose internal parameters were learned from data, rather than hand-coded.

Parameters (Weights)

The internal numbers a model adjusts during training — often billions of them.

Training

The process of repeatedly adjusting a model's weights using example data.

Inference

Using an already-trained model to produce a new output — this is what happens when you send a prompt.

Token

A chunk of text — often a word or part of a word — that a model reads or generates one piece at a time.

Prompt

The input text or instructions you give a model to shape its output.

Fine-Tuning

Further training an already-trained model on a smaller, more specific dataset.

Dataset

The collection of examples used to train or evaluate a model.